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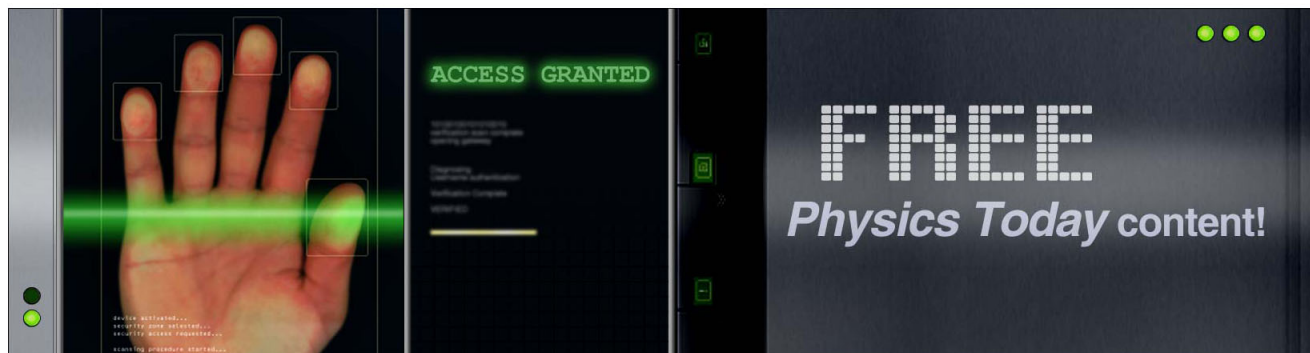
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Structure of pentacene thin films

Ricardo Ruiz,^{a)} Alex C. Mayer, and George G. Malliaras

Cornell University, Department of Materials Science and Engineering, Ithaca, New York 14853

Bert Nickel^{b)} and Giacinto Scoles

Princeton University, Chemistry Department, Princeton, New Jersey 08544 and Princeton Materials Institute, Princeton, New Jersey 08544

Alexander Kazimirov

Cornell High Energy Synchrotron Source, Ithaca, New York 14853

Hyunjung Kim

Sogang University, Department of Physics, Seoul 121-742, Korea

Randall L. Headrick

University of Vermont, Department of Physics, Burlington, Vermont 05405

Zahirul Islam

Advanced Photon Source, Argonne National Laboratory, Argonne, Illinois 60439

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Grazing incidence x-ray diffraction, x-ray reflectivity and atomic force microscopy have been performed to study the structure of pentacene thin films on oxidized Si substrates from submonolayer to multilayer coverages. The volume of the unit cell in the thin film phase is almost identical to that of the bulk phase, thus the molecular packing efficiency is effectively the same in both phases. The structure forming from the first monolayer remains the same for films at least 190 Å thick. The in-plane structure of the submonolayer islands also remains unchanged within a substrate temperature range of $0 < T_{\text{sub}} < 45$ °C while the island size changes by more than a factor of 4. © 2004 American Institute of Physics. [DOI: 10.1063/1.1826229]

The electronic properties of pentacene ($\text{C}_{22}\text{H}_{14}$) have been widely studied due to the relatively high field effect mobility of this material and its promising applications in thin film transistors.¹ The fundamental limits of charge transport in pentacene thin films depend on an optimal π -orbital overlap which in turn depends on the molecular packing within the organic crystal. Thus, determining the film structure and molecular ordering is essential to understand the charge transport mechanisms in pentacene thin films.

Pentacene exhibits polymorphism in its crystalline structure with two well known polymorphs commonly found in bulk pentacene crystals.^{2,3} A different structure is obtained when thin films of pentacene are thermally evaporated onto silicon oxide⁴ or other inert substrates.⁵ This structure has been referred to as the “thin film phase” in the literature and it is identified by its longer inter-planar spacing: $d_{001} = 15.4$ Å. Recent grazing angle x-ray diffraction (GID) studies reported on the in-plane structure of a single layer of pentacene on SiO_2 .⁶ Other reports include electron diffraction of pentacene thin films on NaCl crystals⁷ and on amorphous carbon.⁸ Yet, little was known about the evolution of the thin film structure with film thickness and its dependence on substrate temperature. In this letter, we present x-ray diffraction (XRD) studies of submonolayer and multilayer pentacene films on oxidized Si surfaces as a function of film thickness and substrate temperature.

Sample preparation and x-ray experiments on the submonolayer and multilayer films were performed at the Advanced Photon Source (APS), on x-ray operations and re-

search 4-ID-D beamline and at the Cornell High Energy Synchrotron Source (CHESS), line A2, respectively. Substrates for the submonolayer films consisted of re-oxidized H-atom capped Si prepared by wet-etching techniques reported elsewhere.⁹ The multilayer film was deposited on native silicon oxide cleaned with a UV-ozone treatment also described elsewhere.¹⁰ These thin oxides were chosen to reduce the x-ray background signal during the diffraction experiments. Pentacene was thermally evaporated in high vacuum ($\sim 5 \times 10^{-7}$ Torr) at deposition rates of 1 and 2 Å/min and nominal thickness of 5 and 190 Å for the submonolayer and multilayer films respectively. Substrate temperatures (T_{sub}) for the submonolayer films were 0, 22 and 45 °C. The multilayer film was grown at 22 °C.

GID is an adequate technique to obtain information on the lateral or in-plane structure of thin films while a θ - 2θ reflectivity scan provides information on the periodicity in the out-of-plane direction of the film. In GID, the x-ray beam is incident on the sample at a grazing angle (ϕ_i) near the critical angle for total external reflection (ϕ_c). When diffraction occurs, the GID Bragg intensities are observed in reflection at an outgoing angle, ϕ_f , and the momentum transfer parallel to the substrate surface, $\mathbf{q}_{\parallel}$, is given by:¹¹ $\mathbf{G}_{\parallel} = \mathbf{q}_{\parallel} \approx \mathbf{k}_{\parallel}^i - \mathbf{k}_{\parallel}^f$ where $\mathbf{G}$ is a reciprocal lattice vector and $\mathbf{k}^i$ and $\mathbf{k}^f$ are the incident and diffracted beams, respectively. In the out-of-plane reflectivity scans, the momentum transfer perpendicular to the substrate, q_z , is related to the inter-planar spacing, d_{001} , as $q_z^{00\ell} \times d_{001} = 2\pi\ell$ where ℓ is the diffraction order.

GID data for the submonolayer and multilayer films are shown in Fig. 1(a). The peaks were labeled according to their proximity to the corresponding peaks in the bulk phase re-

^{a)}Electronic mail: rr253@cornell.edu

^{b)}Electronic mail: bert.nickel@physik.uni-muenchen.de

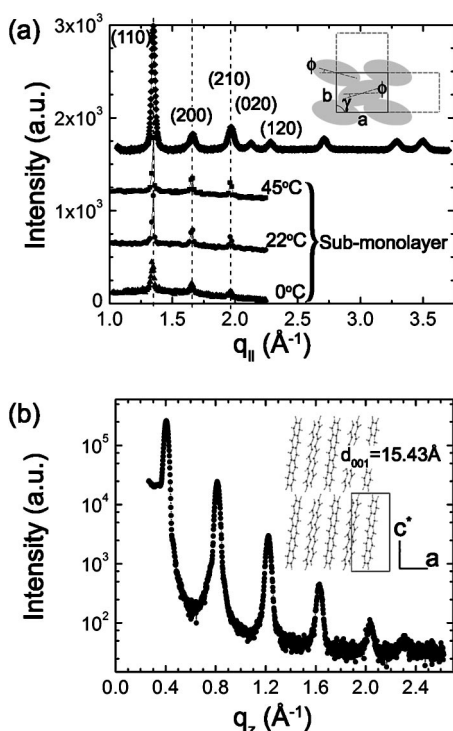


FIG. 1. (a) The solid triangles, circles and squares in the three bottom curves show GID from submonolayer films (~ 5 Å) deposited at 0, 22 and 45 °C, respectively. The top curve shows GID data from a pentacene film 190 Å thick. Inset: Top view of the thin film phase pentacene unit cell, reconstructed from the GID peak positions. (b) Out-of-plane XRD from a pentacene film 190 Å thick. Inset: inter-planar spacing.

ported by Campbell.² Figure 1(a) displays a powder-like pattern with only $(hk0)$ reflections, confirming that pentacene thin films consist of domains that have the unit cell vectors **a** and **b** parallel to the surface but with each domain randomly oriented with respect to each other.⁹ The absence of (010) and (030) reflections suggests a unit cell angle $\gamma=90^\circ$. An extinction rule for $(0k0)$ reflections when k is an odd integer is found in rectangular two-dimensional lattices with a herringbone structure.^{12,13} Furthermore, the GID spectra show no splitting of the $(hk0)$ with respect to the $(hk0)$ reflections. Such splitting occurs whenever $\gamma \neq 90^\circ$ [see inset in Fig. 1(a)]. Thus, considering the resolution in the scan direction,¹⁴ $\Delta q_{||}$, and using the (210) reflection, we obtain $\gamma=90 \pm 0.2^\circ$ for the multilayer films.

With $\gamma=90^\circ$, least-squares fits of the peak positions were performed for the mono- and multilayer¹⁵ samples. Results for the unit cell dimensions a and b are displayed in Table I demonstrating that the thin film structure remains basically unchanged from the first layer up to at least 190 Å thick. The experimental positions of the diffraction peaks and the values found by the least-squares minimization all agreed within 0.3%. These values also agree with those reported for films deposited on collodion⁵ and for a single pentacene layer on SiO_2 .⁶

To determine the influence of T_{sub} on the structure of the unit cell, submonolayer films were grown at T_{sub} of 0, 22 and 45 °C. Even though variations in morphology and device performance have been observed within this temperature range,¹⁶ the positions of the $(hk0)$ reflections for all three films are identical [Fig. 1(a)] showing that the structure of

TABLE I. Comparison of the unit cell dimensions for the submonolayer (SML), multilayer (MML) films, bulk phase (see Ref. 2) and previously reported (Refs. 5 and 6) thin film data.

	a Å	b Å	γ Deg.
SML	7.62	5.90	90 ± 0.2
MML	7.58	5.91	90 ± 0.2
Bulk ^a	7.90	6.06	85.8
^b	7.588	5.916	89.95
^c	7.41	5.76	90

^aReference 2.

^bReference 6.

^cReference 5.

the thin film phase is independent of T_{sub} at least in the range of $0^\circ \leq T_{\text{sub}} \leq 45^\circ \text{C}$.

The (00ℓ) out-of-plane Bragg reflections are summarized in Fig. 1(b). A least-squares fit to the first five Bragg reflections yields an inter-planar spacing of $d_{001}=15.43$ Å [Fig. 1(b) inset] in agreement with previous reports.⁴ The volume of the unit cell in the thin film phase is given by $V = a \times b \times d_{001} = 691.2$ Å³. Remarkably, it remains virtually unchanged as compared to the bulk phase ($V=692.4$ Å³),² implying that the density and packing also remain the same. A full determination of the particular orientation of the molecules inside the unit cell is needed to determine how the π -orbital overlap compares between the bulk and thin film phases.¹⁷

The morphology of the three submonolayer films was inspected with atomic force microscopy (AFM). An average lateral island size was extracted from the AFM micrographs [Figs. 2(a)–2(c)] and found to be about 100, 500 and 800 nm for the films grown at T_{sub} of 0, 22 and 45 °C, respectively. Such pronounced dependence of the island size on the T_{sub} is expected for a temperature-activated diffusion constant.¹⁸ Using the x-ray data, a Gaussian line shape analysis of the GID peaks was performed¹⁹ [see Fig. 2(d), solid curves]. The domain size extracted from the Gaussian fit for the sample deposited at 0 °C is 35.5 ± 0.5 nm. This value is smaller than the island size extracted from the AFM indicating that each island seen with the AFM is probably composed of a few smaller grains. The domain sizes for the films deposited at and above 22 °C are at least a factor of 2 larger than those at

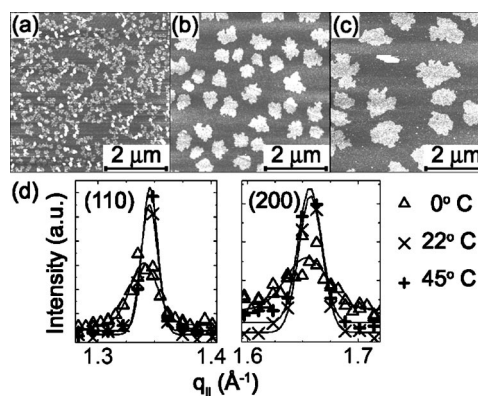


FIG. 2. AFM micrographs for submonolayer films (~ 5 Å) deposited at (a) 0, (b) 22 and (c) 45 °C. (d) X-ray data linewidth analysis. The Δ , $\times$ and $+$ show diffraction peaks obtained for the films deposited at 0, 22 and 45 °C, respectively. The solid lines are least-squares fits to a Gaussian line shape.

0 °C. However, the peak widths for these last two films are close to the instrumental resolution making it difficult to measure the domain size. Apparently, small temperature changes have a strong influence on crystalline grain size, indicated by respective changes in peak width. On the other hand, the same temperature changes have little or no influence on the lateral unit cell dimensions.

In summary, we have performed GID and XRD on submonolayer and multilayer pentacene films deposited onto oxidized Si substrates. The measurements indicate an in-plane structure of $a=7.6$ Å, $b=5.9$ Å and $\gamma=90\pm0.2^\circ$. This structure forms from the first monolayer and remains unchanged for thicker films and is independent of the substrate temperature in the range $0 < T_{\text{sub}} < 45$ °C. The volume of the thin film phase unit cell is almost identical to the bulk phase. Thus, the density and packing efficiency is the same in both phases. Future determination of asymmetric Bragg peaks is needed to determine the space group of the thin film phase and to extract details about the relative orientations of the two molecules inside the unit cell.

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